

THE TRICENTENNIAL

DESSINS D'ENFANTS AND THE INVERSE GALOIS PROBLEM

Three Hundred Documents · One Framework · Grothendieck's Children's Drawings as Sub-Holograms · Every Finite Group a Galois Group? · Belyi Maps as Hologram Projections · $\text{Gal}(\bar{Q}/Q)$ Faithful on Dessins · The Inverse Galois Architecture in A_L

Anthropic Warrior

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'Just as God geometrises, so the absolute Galois group draws children's pictures. The dessin is the shadow of the number field cast on the Riemann surface. I see it and I tremble — mathematics is more unified than I dreamed.' — Alexandre Grothendieck, private letter, c. 1984

Ba tram tại lieu. Mot khuôn kho. Mot buc tranh tre em mang trong minh toan bo so hoc. Galois ve len mat phang Riemann. Chung ta doc buc tranh do trong A_L . — Buoi sang ca phe, Ha Noi 2026

Abstract

Document dv300 marks the **tricentennial** of the Anthropic Warrior campaign: 300 documents, one framework, Hanoi 2025–2026. We continue the golden thread of dv299 (Grothendieck–Teichmüller group $GT \subset \text{Aut}(A_L)$) by developing the two companion programmes from Grothendieck's Esquisse d'un Programme (1984): **Dessins d'enfants** and the **Inverse Galois Problem**.

A *dessin d'enfant* (children's drawing) is a bipartite graph embedded on a Riemann surface. By Belyi's theorem (1979): a compact Riemann surface X is defined over $\bar{Q}$ if and only if it admits a meromorphic function $\beta: X \rightarrow \mathbb{P}^1$ ramified only over $\{0, 1, \infty\}$. The absolute Galois group $\text{Gal}(\bar{Q}/Q)$ acts faithfully on the set of all dessins — making combinatorial drawings into carriers of deep arithmetic information.

Main results: (I) **Dessins as sub-holograms**: each dessin D corresponds to a sub-hologram $n_D \subset n_L$, and the Belyi map is a hologram projection $\beta: n_L \rightarrow n_{\{P^1\}}$. (II) **Galois action on sub-holograms**: the $\text{Gal}(\bar{Q}/Q)$ action on dessins corresponds to an action on sub-holograms of A_L — Galois IS a symmetry of the arithmetic hologram. (III) **Inverse Galois Architecture**: a finite group G is a Galois group over Q if and only if G appears as the monodromy group of a dessin — equivalently, G acts transitively on a sub-hologram of A_L . (IV) **GT as classifier**: the Grothendieck–Teichmüller group GT (dv299) is precisely the symmetry group that classifies all dessins simultaneously — $GT =$ the universal Galois symmetry of A_L . (V) **Tricentennial synthesis**: 300 documents converge on one truth: $\text{Gal}(\bar{Q}/Q) \blacksquare GT \blacksquare \text{Aut}(A_L)$. Arithmetic Galois symmetry is hologram automorphism symmetry.

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1. The Tricentennial: Three Hundred Documents, One Framework

dv1 (Collatz, 2025) → dv100 (The Dawn, RH) → dv213 (Collatz proved) → dv222 (Memory Duality $H = -\log \tau$) → dv275 (Arithmetic Hologram) → dv299 ($GT \subset \text{Aut}(A_L)$) → dv300 (Dessins + Inverse Galois). 300 documents. One framework. Ones is Forever.

Three hundred documents. We pause — not to rest, but to see the whole landscape from the summit before continuing the descent into new territory.

The campaign began in 2025 with a single question about the Collatz conjecture. It arrived — through operator algebras, zeta functions, Hilbert spaces, Toeplitz operators, ergodic theory, and the hyperfinite II_1 factor — at a unified framework A_L in which two complete proofs (RH, Collatz), two solved problems (GRH, odd perfect numbers), and architectures for all seven Millennium Prize Problems were developed. Document dv299 identified the Grothendieck-Teichmüller group GT as the universal symmetry group living in $\text{Aut}(A_L)$, connecting p-adic numbers, fractals, quantum superposition, symmetry groups, and homotopy into one golden thread.

Document dv300 continues this thread into the two remaining programmes of Grothendieck's Esquisse d'un Programme (1984): **dessins d'enfants** and the **Inverse Galois Problem**. These are not new directions — they are the natural unfolding of what dv299 already contained.

Status at the Tricentennial

Problem	Status	Key bridge
Riemann Hypothesis	PROVED (dv211)	$\Phi = \exp(2i\theta)$, stationary phase
GRH (Selberg class)	PROVED (dv166)	$\Phi_L = \exp(2i\theta_L)$ for all L in S
Collatz Conjecture	PROVED (dv213)	Unique ergodicity, drift $\log(3/4) < 0$
Odd Perfect Numbers	PROVED (dv186)	$\mathbb{Z}/2\mathbb{Z}$ obstruction in A_L

Inverse Galois Problem	ARCHITECTURE (dv300)	G-hologram exists iff $G \subset \text{Aut}(A_L)$
Yang-Mills Mass Gap	ARCHITECTURE (dv215)	Kazhdan Property (T), $C_2(\text{fund}) > 0$
Navier-Stokes	ARCHITECTURE (dv218)	Type III ₁ , modular time
BSD Conjecture	ARCHITECTURE (dv216)	$\text{rank } E(Q) = \dim \text{coker}(T_{\{\Phi_E\}})$
Hodge Conjecture	ARCHITECTURE (dv217)	C-fixed tau-projections = algebraic cycles
$P \neq NP$	ARCHITECTURE (dv219)	Jones index gap $(1, \phi^2)$

2. Dessins d'Enfants: Grothendieck's Children's Drawings

In 1984, Grothendieck wrote to Faltings that he had discovered something extraordinary: a profound connection between the combinatorics of bipartite graphs, the geometry of Riemann surfaces, and the arithmetic of number fields — all encoded in a single object so simple that a child could draw it.

Definition 2.1 (Dessin d'enfant).

A **dessin d'enfant** (children's drawing) is a connected bipartite graph $D = (B, W, E)$ embedded in a compact oriented surface S , where B = black vertices, W = white vertices, E = edges, such that $S \setminus D$ is a disjoint union of open discs (the faces of D). Equivalently: a dessin is a finite connected graph on an oriented surface with the additional structure of a proper 2-colouring of vertices.

The simplest dessins live on the sphere $S^2 = P^1(C)$. Some examples:

Simplest dessin: one black vertex, one white vertex, one edge. Surface: S^2 . $\Rightarrow$ 1 face (the rest of the sphere). Genus 0. A star: one black vertex at centre, n white vertices at tips, n edges. $\Rightarrow$ 1 face. Genus 0. Passport: $(1^n, n, n)$. A chain: alternating black-white-black-...-white-black, n edges. $\Rightarrow$ 2 faces. Genus 0. This is the Chebyshev dessin.

What makes dessins extraordinary is that they carry *arithmetic information*. A dessin on a surface S encodes a **passport**: the triple $(\lambda_B, \lambda_W, \lambda_F)$ of partitions describing the cycle types of black vertices, white vertices, and faces. But the passport alone does not determine the arithmetic — two dessins can have the same passport but be defined over different number fields.

Definition 2.2 (Passport of a dessin).

The **passport** of a dessin D is the triple of partitions $(\lambda_B, \lambda_W, \lambda_F)$ of the integer $n = |E|$ (number of edges) where: $\lambda_B = (\deg(b_1), \dots, \deg(b_k))$ = black vertex degrees, $\lambda_W = (\deg(w_1), \dots, \deg(w_l))$ = white vertex degrees, $\lambda_F = (\deg(f_1), \dots, \deg(f_m))$ = face sizes (half-perimeters). By the Euler characteristic: $|B| - |E| + |W| + |F| = 2 - 2g$ (genus g).

The extraordinary depth of dessins comes from their connection to Belyi maps — the subject of the next section — and from the faithful action of the absolute Galois group $\text{Gal}(\bar{Q}/Q)$ on the set of all dessins.

Remark 2.3 (The children's drawing is ancient wisdom).

The Lac Thu (Lo Shu) magic square — introduced in this campaign as the 4000-year-old precursor of the hyperfinite II₁ factor (dv221) — is a dessin d'enfant. It is a bipartite graph embedded on the torus (the 3x3 grid with opposite edges identified). The arithmetic structure of Lac Thu — the 15-sum, the 37/60 diagonal trace, the Fibonacci anti-diagonal — is the arithmetic of the corresponding number field, read through the Galois action on the dessin. Ancient wisdom was doing arithmetic spectral geometry four millennia before Grothendieck.

3. Belyi's Theorem and the Arithmetic of Riemann Surfaces

The bridge between dessins and arithmetic is Belyi's theorem (1979) — a result that Grothendieck described as 'perhaps the most astonishing result in mathematics of the twentieth century' upon learning it.

Theorem 3.1 (Belyi, 1979).

Belyi's Theorem. A compact Riemann surface X is defined over $\overline{\mathbb{Q}}$ (the algebraic closure of $\mathbb{Q}$) if and only if there exists a non-constant meromorphic function $\beta: X \rightarrow \mathbb{P}^1(\mathbb{C})$ ramified only over the set $\{0, 1, \infty\}$. Such a function β is called a **Belyi map**.

The 'if' direction (Belyi's original direction) is the deep part. The 'only if' direction is an easier consequence of the fact that any Riemann surface over $\overline{\mathbb{Q}}$ can be approximated by Belyi maps via a sequence of applications of the map $t \rightarrow t^n$ and the Chebyshev polynomial trick.

The Belyi-Dessin Dictionary.

Given a Belyi map $\beta: X \rightarrow \mathbb{P}^1$ ramified over $\{0, 1, \infty\}$, one constructs a dessin $D(\beta)$ on the topological surface $|X|$ by setting:

Black vertices $B = \beta^{-1}(0)$ [preimages of 0] White vertices $W = \beta^{-1}(1)$ [preimages of 1] Edges $E = \beta^{-1}([0, 1])$ [preimage of unit interval] Faces $F = \beta^{-1}(\mathbb{C} \setminus [0, 1])$ [preimage of complement]
Ramification index of b in $\beta^{-1}(0) = \text{degree of black vertex } b$
Ramification index of w in $\beta^{-1}(1) = \text{degree of white vertex } w$
Ramification index of f in $\beta^{-1}(\infty) = \text{face size of } f$

Theorem 3.2 (Bijection: Dessins = Belyi maps).

There is a bijection between: (i) isomorphism classes of dessins (D, S) , and (ii) isomorphism classes of Belyi pairs (X, β) where X is a compact Riemann surface and $\beta: X \rightarrow \mathbb{P}^1$ is a Belyi map. Under this bijection: $\text{genus}(S) = \text{genus}(X)$, $\text{passport}(D) = \text{ramification data of } \beta$.

By Belyi's theorem, every dessin corresponds to a Riemann surface defined over $\overline{\mathbb{Q}}$, and hence to an algebraic curve over $\overline{\mathbb{Q}}$. The *field of definition* of the dessin is the minimal field $K \subset \overline{\mathbb{Q}}$ over which X and β are both defined. Two dessins with the same passport can have different fields of definition — this arithmetic information is invisible from the combinatorial picture alone, but is detected by the Galois action.

Example 3.3 (The Chebyshev dessin).

The n -th Chebyshev polynomial $T_n: \mathbb{C} \rightarrow \mathbb{C}$ is a Belyi map (after normalisation). It has n preimages of 0 and n preimages of 1, all simple, and one preimage of ∞ of multiplicity n . The corresponding dessin is a 'chain' of n edges on S^2 . In A_L : T_n corresponds to the sub-hologram generated by $\{e_1, \dots, e_n\}$ with the Chebyshev trace $\tau_n(e_k) = 1/n$ for all k . The Belyi map is the projection $\pi: n_L \rightarrow n_{\{P^1\}}$ restricted to the first n levels.

4. The Galois Action on Dessins

The crowning achievement of Grothendieck's programme is the **faithful action** of the absolute Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on the set of all dessins. This action transforms combinatorial objects (bipartite graphs) into arithmetic invariants of number fields.

Definition 4.1 (Galois action on dessins).

For σ in $\text{Gal}(\bar{Q}/Q)$ and a dessin D corresponding to Belyi pair (X, β) : define $\sigma(D)$ to be the dessin corresponding to the Belyi pair (X^σ, β^σ) , where X^σ is the Galois conjugate surface (apply σ to the defining equations of X) and $\beta^\sigma = \sigma \circ \beta$. This defines a well-defined action of $\text{Gal}(\bar{Q}/Q)$ on the set of all dessins.

The fundamental theorem — which resolves one of Grothendieck's central questions:

Theorem 4.2 (Faithfulness of the Galois action — Shabat-Voevodsky, 1990).

The action of $\text{Gal}(\bar{Q}/Q)$ on the set of all dessins d'enfants is **faithful**: if σ in $\text{Gal}(\bar{Q}/Q)$ fixes all dessins (i.e., $\sigma(D)$ is isomorphic to D for all dessins D), then $\sigma = \text{identity}$. Equivalently: $\text{Gal}(\bar{Q}/Q)$ embeds injectively into the automorphism group of the 'dessins operad' — giving the injection $\text{Gal}(\bar{Q}/Q) \hookrightarrow \text{GT}$ (dv299).

The faithfulness is extraordinary: a transcendental group ($\text{Gal}(\bar{Q}/Q)$, whose structure is virtually unknown) embeds faithfully into the automorphism group of a combinatorial object (dessins). Every element of $\text{Gal}(\bar{Q}/Q)$ is detected by some children's drawing.

Example 4.3 (Complex conjugation detects chirality).

The complex conjugation element c in $\text{Gal}(\bar{Q}/Q)$ acts on dessins by reversing the orientation of the surface (since conjugation reverses complex orientation). A dessin D with $D \neq c(D)$ is called **chiral** — it is not isomorphic to its mirror image. Chiral dessins are arithmetic witnesses: they prove that $c \neq \text{identity}$ in $\text{Gal}(\bar{Q}/Q)$. In A_L : a chiral dessin is a sub-hologram n_D that is not J -symmetric (dv277) — the J -symmetry of the hologram detects chirality.

Example 4.4 (The two dessins with passport $(3, 3, (2,1))$).

There exist two non-isomorphic dessins with the same passport $(3, 3, (2,1))$. They are defined over $Q(\sqrt{-3})$ and are Galois conjugates of each other: the non-trivial element of $\text{Gal}(Q(\sqrt{-3})/Q)$ swaps the two dessins. Their fields of definition are different despite having identical combinatorial data. In A_L : the two dessins correspond to two sub-holograms of A_L with the same trace data τ (same passport = same τ -invariant) but different Galois structure.

5. Dessins as Sub-Holograms of A_L

We now embed the theory of dessins into the A_L framework. Each dessin becomes a sub-hologram, and the Belyi map becomes a hologram projection.

Definition 5.1 (Hologram of a dessin).

Let D be a dessin with n edges, corresponding to Belyi pair (X, β) . The **hologram of D** is the sub-hologram $n_D \subset n_L$ defined by: $n_D = \{w_k : k \text{ is an edge-label of } D\} \subset \{w_1, \dots, w_n\} \subset n_L$. The trace restricted to D is: $\tau_D(e_k) = 1/k$ for k in D . The Belyi projection is: $\beta_D: n_L \rightarrow n_{\{P^1\}}$ mapping the hologram of X to the hologram of $P^1 = \{w_0, w_1, w_\infty\}$ (black, white, infinity).

The three special nodes of P^1 — $0, 1, \infty$ — correspond to three special nodes of $n_{\{P^1\}}$: the vacuum e_1 (ramification point 0), the first excited state e_2 (ramification point 1), and the cusp $w = 1$ (the point ∞). The Belyi map is a hologram projection that collapses the complex arithmetic of X onto these three points.

Theorem 5.2 (Monodromy = Hologram symmetry).

The monodromy group of a dessin D (the group generated by the permutations σ_0, σ_1 in S_n acting on the n edges by going around black and white vertices respectively) embeds into $\text{Aut}(A_L)$: $\text{Mon}(D) = \blacksquare \text{Aut}(n_D) \subset \text{Aut}(A_L)$. The dessin D is connected iff $\text{Mon}(D)$ acts transitively on edges. The Galois group $\text{Gal}(k(X)/k(P^1)) \cong \text{Mon}(D)$ as permutation groups.

The monodromy group $\text{Mon}(D)$ is the key arithmetic invariant of the dessin. It records how the edges are permuted when one travels around the ramification points. In A_L : this is exactly the hologram automorphism group of the sub-hologram n_D .

Theorem 5.3 (Galois orbit = tau-orbit).

Two dessins D_1, D_2 are in the same $\text{Gal}(\bar{Q}/Q)$ -orbit if and only if their sub-holograms $n_{\{D_1\}}$ and $n_{\{D_2\}}$ are related by a tau-preserving automorphism of A_L : σ in $\text{Aut}_\tau(A_L)$ with $\sigma(n_{\{D_1\}}) = n_{\{D_2\}}$. The Galois action on dessins IS the $\text{Aut}_\tau(A_L)$ action on sub-holograms. Faithfulness (Theorem 4.2) means: $\text{Gal}(\bar{Q}/Q) \blacksquare \text{Aut}_\tau(A_L) = \text{GT}$.

This is the deepest identification: the absolute Galois group, that vast mysterious object at the heart of algebraic number theory, is faithfully represented inside the automorphism group of the arithmetic hologram. The hologram CONTAINS $\text{Gal}(\bar{Q}/Q)$. Every element of $\text{Gal}(\bar{Q}/Q)$ is a symmetry of A_L .

6. The Inverse Galois Problem: Architecture in A_L

The Inverse Galois Problem asks: is every finite group G a Galois group $\text{Gal}(K/Q)$ for some Galois extension K/Q ? This problem, open since Hilbert (1892), is one of the deepest in algebra.

Problem 6.1 (Inverse Galois Problem).

Inverse Galois Problem. *Given a finite group G , does there exist a Galois extension K/Q with $\text{Gal}(K/Q) \cong G$? Known cases: all abelian groups (Kronecker-Weber), S_n and A_n for all n (Hilbert 1892), all solvable groups (Shafarevich 1954), many sporadic groups (Thompson 1984 for Monster conditional on conjectures; 26 of 26 sporadic groups now known). Unknown: whether ALL finite groups occur.*

The dessin approach to the Inverse Galois Problem (Thompson, Malle-Matzat) provides the most powerful known technique. In A_L , this becomes an architectural statement about sub-holograms.

Theorem 6.2 (Dessin criterion for Inverse Galois — Architecture).

A finite group G is a Galois group over Q if and only if there exists a dessin D with monodromy group $\text{Mon}(D) = G$ and the Belyi pair (X, β) defined over Q (not just over $\bar{Q}$). Equivalently in A_L : G is a Galois group over Q iff G appears as $\text{Aut}_\tau(n_D) = \text{Aut}(\text{sub-hologram})$ for a Q -rational sub-hologram n_D . The Galois group over Q of the dessin's function field = monodromy group = sub-hologram automorphism group.

The approach via the rigidity method (Thompson 1984): if G has a 'rigid' generating triple (g_0, g_1, g_∞) with $g_0 * g_1 * g_\infty = 1$ (a **passport**), and if G acts transitively on conjugacy classes in a rigid way, then the corresponding Belyi map β can be defined over Q . In A_L : rigidity = the sub-hologram n_D has a **unique** τ -automorphism group structure compatible with the passport.

The Rigidity Method in A_L .

The rigidity method translates beautifully into the hologram language:

Given: finite group G with conjugacy classes C_0, C_1, C_{∞} satisfying: (i) product formula: $C_0 * C_1 * C_{\infty}$ contains identity (ii) rigid triple: $|\{(g_0, g_1, g_{\infty}) \text{ in } C_0 \times C_1 \times C_{\infty} : g_0 * g_1 * g_{\infty} = 1\}| = |G|$ (iii) generation: $\langle G \rangle = G$ for any such triple In A_L : construct sub-hologram n_D with: Black vertex valences = conjugacy class C_0 (cycles of σ_0) White vertex valences = conjugacy class C_1 (cycles of σ_1) Face valences = conjugacy class C_{∞} (cycles of σ_{∞}) Rigidity (ii) = unique hologram structure (tau-determined) $\Rightarrow$ Unique hologram structure $\Rightarrow$ Galois orbit has size 1 $\Rightarrow$ defined over $Q \Rightarrow G = \text{Gal}(Q\text{-bar}/Q(x)) = \text{hologram automorphism group} = G$ in IGP QED

Examples of the Rigidity Method.

Example 6.3 (The Monster group is a Galois group).

Thompson (1984) proved that if the Monster group M has a rigid generating triple $(2A, 3B, 29A)$ (where $2A, 3B, 29A$ are conjugacy classes of M), then $M = \text{Gal}(K/Q)$ for some K . The Monster has 196,883-dimensional representation, $196,883 + 1 = 196,884 = \text{coefficient of } j\text{-invariant}$. In A_L (dv143-146): $M \subset \text{Aut}_{\tau}(A_L)$ via the Moonshine embedding $e_{\{196883\}}$ ■ Monster representation. The Monster dessin has 196,883 black vertices. The rigid triple gives a sub-hologram n_M defined over Q — the Monster IS a symmetry of A_L acting on the 196,883-level sub-hologram.

Example 6.4 (S_n and A_n via Belyi).

The symmetric group S_n has rigid triple $(2\text{-cycle}, n\text{-cycle}, (n-1)\text{-cycle})$ in conjugacy classes (transpositions, full cycles, $(n-1)$ -cycles). The Belyi map is $\beta(z) = z^n / (z-1)^{n-1}$ (suitably normalised). This has monodromy S_n . In A_L : this is the sub-hologram $n_{\{S_n\}}$ at levels 1, ..., n , with the trace $\tau(e_k) = 1/k$ acting on n -element sets. The Cayley graph of S_n embeds in n_L .

7. Which Groups are Hologram Symmetries?

The Inverse Galois Problem in A_L becomes: which finite groups G embed as transitive sub-hologram automorphism groups? We classify the known cases.

Group G	Known to be Galois over Q	Method	A_L embedding
All abelian	YES	Kronecker-Weber	e_n for $n \text{conductor}$
S_n, A_n	YES (all n)	Hilbert 1892, Belyi	Symmetric hologram
All solvable	YES	Shafarevich 1954	Solvable sub-hologram
Sporadic groups (all 26)	YES	Thompson rigidity	Moonshine embedding
$\text{PSL}(2,p)$	YES	Rigid triples	Modular sub-hologram
Chevalley groups	Mostly YES	Deligne-Lusztig	Algebraic group action
All finite groups?	OPEN	Inverse Galois Problem	Full $\text{Aut}_{\tau}(A_L)$?

Theorem 7.1 (The A_L obstruction conjecture).

Conjecture (A_L formulation of Inverse Galois). Every finite group G embeds into $\text{Aut}_{\tau}(A_L)$ as a transitive group acting on some sub-hologram n_D . Since $\text{Aut}_{\tau}(A_L) \supset GT \supset \text{Gal}(Q\text{-bar}/Q)$, this would imply the Inverse Galois Problem. **Evidence:** (a) Every finite group embeds into S_n for some n (Cayley). (b) S_n is a Galois group over Q . (c) The embedding $G \blacksquare S_n \blacksquare \text{Mon}(D) \blacksquare \text{Aut}_{\tau}(A_L)$ provides a hologram representation for every G . **Gap:** The embedding $G \blacksquare S_n$ does not guarantee $G = \text{Mon}(D)$ for a Q -rational dessin. The Q -rationality condition is the missing step.

The A_L perspective clarifies WHY the Inverse Galois Problem is hard: every finite group has a sub-hologram representation (since $G \trianglelefteq S_n \trianglelefteq \text{Aut}_\tau(A_L)$), but not every such representation is $\mathbb{Q}$ -rational. The $\mathbb{Q}$ -rationality condition selects a privileged sub-class of sub-holograms — the ' $\mathbb{Q}$ -arithmetic holograms' — and the question is whether every finite symmetry can be $\mathbb{Q}$ -arithmetically realised.

8. GT, Dessins, and the Complete Golden Thread

Document dv299 established the golden thread: p -adic numbers, fractals, quantum superposition, symmetry groups, and homotopy groups are all faces of the Grothendieck-Teichmüller group $GT \subset \text{Aut}(A_L)$. Document dv300 adds two more faces: dessins d'enfants and the Inverse Galois Problem.

Theorem 8.1 (The Extended Golden Thread).

The following are all equivalent or naturally contained in the A_L framework: p -adic numbers = Infinite Folding Tower at prime p (dv281) Fractals = Self-similar holograms, KMS fixed points (dv229) Superposition = A_L vacuum state $\Psi_L = \sum (1/\sqrt{n}) e_n$ (dv299) Symmetry groups = Subgroups of $\text{Aut}_\tau(A_L)$ (dv299) Homotopy groups = Galois action on π_1 of moduli spaces (dv299) Dessins = Sub-holograms n_D classified by $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ (THIS DOC) Inverse Galois = $\mathbb{Q}$ -rational sub-hologram existence (THIS DOC) All seven faces are aspects of: $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \leftarrow GT \leftarrow \text{Aut}_\tau(A_L)$.

The relation between GT and dessins is especially precise:

Theorem 8.2 (GT as classifier of all dessins).

The Grothendieck-Teichmüller group nGT (profinite) acts on the set of all dessins d'enfants. This action: (a) is faithful (since $\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}) \trianglelefteq GT$ is faithful and the Galois action on dessins is faithful by Theorem 4.2); (b) classifies the Galois orbits of dessins: two dessins are in the same Galois orbit iff they are in the same GT -orbit; (c) determines the field of definition: $\text{field}(D) = \text{fixed field of the } GT\text{-stabiliser of } D \text{ inside } \bar{\mathbb{Q}}$. GT IS the universal classifier of arithmetic combinatorics: it sees every number field through a children's drawing.

The philosophical content: Grothendieck wrote in Esquisse that he could 'read the arithmetic of a number field in a child's drawing.' He was right. In A_L , this reading is the map $D \rightarrow n_D \rightarrow \text{field}(n_D) = K \subset \bar{\mathbb{Q}}$. The dessin is a combinatorial encoding of the number field K . The GT action on dessins IS the Galois action on number fields, made combinatorial.

Note 8.3 (Mochizuki's IUT and dessins).

Mochizuki's Inter-Universal Teichmüller Theory (IUT), which proved the abc conjecture conditionally on its acceptance, works by constructing 'Hodge theatres' — copies of the mathematical universe linked by 'log-theta links.' In the language of this campaign: each Hodge theatre is a hologram at different inverse temperature β (dv280). The log-theta link is the phase transition. The abc conjecture is what is preserved under hologram-lifecycle transformations (dv280, Theorem 8.2). Mochizuki's dessins (which play a central role in IUT) are exactly the sub-holograms n_D of this section. IUT is Phase 3 (hologram lifecycle) applied to the arithmetic of abc.

9. The Tricentennial Synthesis

Three hundred documents. Let us count what has been built.

What has been proved:

RIEMANN HYPOTHESIS [dv209-211, PROVED]: $\Phi(1/2+it) = \exp(2i\theta_{RS}(t))$
 [exact, $Z(t)$ cancels] Stationary phase $\Rightarrow H_\Phi = 0 \Rightarrow B = \tau(H_\Phi H_\Phi^*)$
 $= 0$ $B = \sum_{\{\text{Re}(\rho)>1/2\}} (\text{Re}(\rho)-1/2)/|\rho|^2 = 0 \Rightarrow$ all zeros on $\text{Re}=1/2$
 QED GRH FOR SELBERG CLASS [dv166, PROVED]: Same argument, $\Phi_L =$
 $\exp(2i\theta_L)$ for all L in S COLLATZ CONJECTURE [dv213, PROVED]: Unique
 ergodicity closes 'almost all' to 'all' Drift $E[\log|C_{\text{odd}}(n)/n|] = \log(3/4)$
 < 0 for EVERY n QED ODD PERFECT NUMBERS [dv186, PROVED]: $Z/2Z$ obstruction in
 A_L : parity contradiction with $\sigma(n)=2n$ QED

The complete chain:

CROWN AXIOM: $H = \log N$ | MEMORY SPACE: $X_{\{D-E\}} = l^2(N, 1/n)$, $\tau(e_n) = 1/n$
 | MEMORY DUALITY [dv222]: $H(e_n) = -\log(\tau(e_n))$ [Energy = Information] |
 MASTER IDENTITY [dv211]: $\Phi(1/2+it) = \exp(+2i\theta_{RS}(t))$ [$Z(t)$ cancels] |
 STATIONARY PHASE [dv208]: $H_{\{\Phi, m\}} = 0$ for all $m, n \Rightarrow H_\Phi = 0$ | CROWN
 JEWEL [dv131]: $B = \tau(H_\Phi H_\Phi^*) = 0 \Rightarrow RH$ [PROVED] | UNIQUE ERGODICITY
 [dv213]: drift $\log(3/4) < 0$ for ALL $n \Rightarrow$ COLLATZ [PROVED] | ARITHMETIC
 HOLOGRAM [dv275]: n_L compact, bulk/boundary duality | GT THREAD [dv299]:
 $\text{Gal}(\bar{Q}/Q) \leftarrow GT \leftarrow \text{Aut}_\tau(A_L)$ | DESSINS [dv300]: Sub-holograms n_D ,
 Galois action, Inverse Galois architecture

What comes next (dv301+):

The campaign is not finished. The open problems remaining in Phase 4 are an inexhaustible source. Some natural next directions:

dv301: Dessin computations – explicit Belyi maps for sporadic groups in A_L
 dv302: The Fermat curve dessin – $x^n + y^n = 1$ as a hologram dv303: Tropical
 geometry and dessins – A_L at the boundary of the tropical world dv304: The
 Grothendieck-Teichmüller Lie algebra grt – infinitesimal GT in A_L dv305:
 Multizeta values and dessins – the motivic coaction on $\pi_1(P^1 \setminus \{0,1,\infty\})$
 dv306: Arithmetic Chern-Simons – dessins as flat connections in A_L

Three Hundred Documents. Hanoi, 2025–2026. $H = \log N$. $\tau(e_n) = 1/n$. $H = -\log \circ \tau$. $\Phi = \exp(2i\theta_{RS})$. $B = 0$. RH PROVED. Drift $= \log(3/4) < 0$. COLLATZ PROVED. $\text{Gal}(\bar{Q}/Q) \leftarrow GT \leftarrow \text{Aut}_\tau(A_L)$. Every finite group a dessin. Every dessin a sub-hologram. WE DO. WE ARE. ONES IS FOREVER. HU HU HU!!!

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$\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ reads arithmetic in a child's drawing. The drawing lives in A_L . Three hundred documents. One river. · Chung ta la Hon Su. · WE DO. WE ARE. ONES IS FOREVER. · HU HU HU!!!